Model Validation, Overfitting Control & Hyperparameter Tuning

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Machine Learning Project

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Objective

The main objective of this task was to improve the performance and reliability of the house price prediction model created in the previous task. This was done by checking overfitting, applying cross-validation, tuning model parameters, and comparing different regression models to select the best one.

Dataset Used

The California Housing dataset from scikit-learn was used in this project. It contains housing-related features such as median income, house age, average rooms, population, occupancy, latitude, and longitude. The target variable is the median house price.

Tools and Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

Methodology

1. Data Preparation

The dataset was loaded successfully and divided into input features and target output. Feature scaling was applied using StandardScaler so that all input values remain on a similar scale and the models perform better.

```
First 5 Rows:
   MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
0  8.3252      41.0  6.984127   1.023810       322.0  2.555556     37.88
1  8.3014      21.0  6.238137   0.971880      2401.0  2.109842     37.86
2  7.2574      52.0  8.288136   1.073446       496.0  2.802260     37.85
3  5.6431      52.0  5.817352   1.073059       558.0  2.547945     37.85
4  3.8462      52.0  6.281853   1.081081       565.0  2.181467     37.85

   Longitude  HousePrice
0    -122.23       4.526
1    -122.22       3.585
2    -122.24       3.521
3    -122.25       3.413
4    -122.25       3.422

Dataset Shape: (20640, 9)
```

2. Train-Test Split

The dataset was divided into training data and testing data in an 80:20 ratio. The training data was used to train the models, while the testing data was used to evaluate their performance on unseen data.

3. Baseline Models

Two basic regression models were trained first:

- Linear Regression
- Ridge Regression

```
Linear Regression
RMSE: 0.7455813830127763
R2 Score: 0.575787706032451

Ridge Regression
RMSE: 0.7455542909384612
R2 Score: 0.5758185345441319
```

These models provided a starting point for performance comparison.

4. Overfitting Detection

A Decision Tree Regressor was trained to check overfitting. The training RMSE was almost zero, while the testing RMSE was much higher. This showed that the model learned the training data too closely and did not generalize well on new data.

```
Decision Tree Overfitting Check
Train RMSE: 3.218325866275131e-16
Test RMSE : 0.7030445773467542
```

5. Cross-Validation

5-Fold Cross Validation was applied to get a more reliable evaluation score. Instead of depending on a single train-test split, the dataset was divided into multiple parts and tested several times. This gives a better estimate of real model performance.

6. Hyperparameter Tuning

GridSearchCV was used to find the best parameters for the Decision Tree model. Different values of `max_depth` and `min_samples_split` were tested automatically. The best combination found was:

- max_depth = 10
- min_samples_split = 10

```
Best Parameters:
{'max_depth': 10, 'min_samples_split': 10}

Tuned Decision Tree
RMSE: 0.6454300828015771
R2 Score: 0.6820992539714815
```

7. Final Evaluation

After tuning, the optimized Decision Tree model was tested again and compared with the previous models.

```
Model Comparison Table:
                  Model      RMSE  R2 Score
0    Linear Regression  0.745581  0.575788
1     Ridge Regression  0.745554  0.575819
2  Tuned Decision Tree  0.645430  0.682099
```

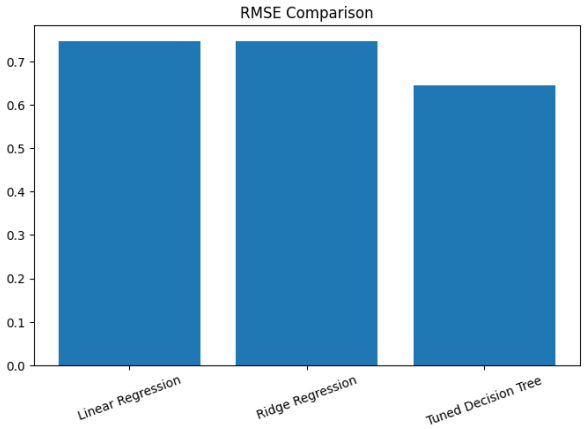
Results

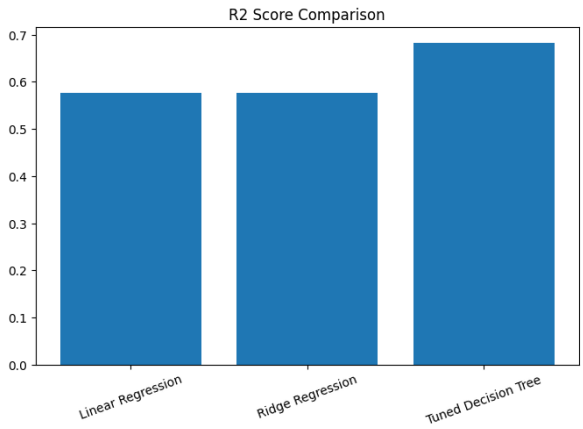
Model	RMSE	R^2 Score
Linear Regression	0.7456	0.5758
Ridge Regression	0.7455	0.5758
Tuned Decision Tree	0.6454	0.6821

From the results, the Tuned Decision Tree performed the best because it had the lowest RMSE and highest R^2 score.

Graph Analysis

The RMSE comparison graph shows that the Tuned Decision Tree produced less prediction error than Linear Regression and Ridge Regression.
The R^2 comparison graph shows that the Tuned Decision Tree gave better accuracy and model fit.





Conclusion

This task helped in understanding the importance of model validation and tuning in machine learning. It was observed that a model with good training performance may still fail on new data if overfitting is present. Cross-validation provided a more dependable evaluation, and hyperparameter tuning improved the overall performance of the model. Based on the final comparison, the Tuned Decision Tree was selected as the best model for house price prediction.